

Global Electronics Retailer: Commercial Performance Audit

Strategic Insights & Recommendations for Sustainable Growth

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Introduction & Business Objectives

Business Context

As operations scale across international markets, identifying core revenue drivers and evaluating supply chain health is critical. This commercial audit aims to uncover purchasing patterns, operational bottlenecks, and channel effectiveness to drive the upcoming fiscal year's growth strategy.

Key Business Questions

1. What is the current composition of our product portfolio, and where are our primary customer bases located?
2. Are there identifiable seasonal patterns impacting revenue and order volume?
3. How is our logistics network performing over time regarding delivery speeds?
4. Is there a measurable difference in the Average Order Value (AOV) between In-Store and Online sales channels?

Data Overview & Preparation

Data Processing (ETL)

Rigorous data cleaning and validation were executed to ensure a high-fidelity analysis of historical global retail transactions.

- **Data Type Casting:** Converted raw string data into exact datetime formats to enable precise time-series analysis.
- **Metric Engineering:** Formulated custom metrics such as Average Order Value (AOV) by aggregating total revenue against distinct orders.

Dimensionality & Tech Stack

Optimized data granularity to focus on macro-level business trends that inform strategic decisions at the executive level.

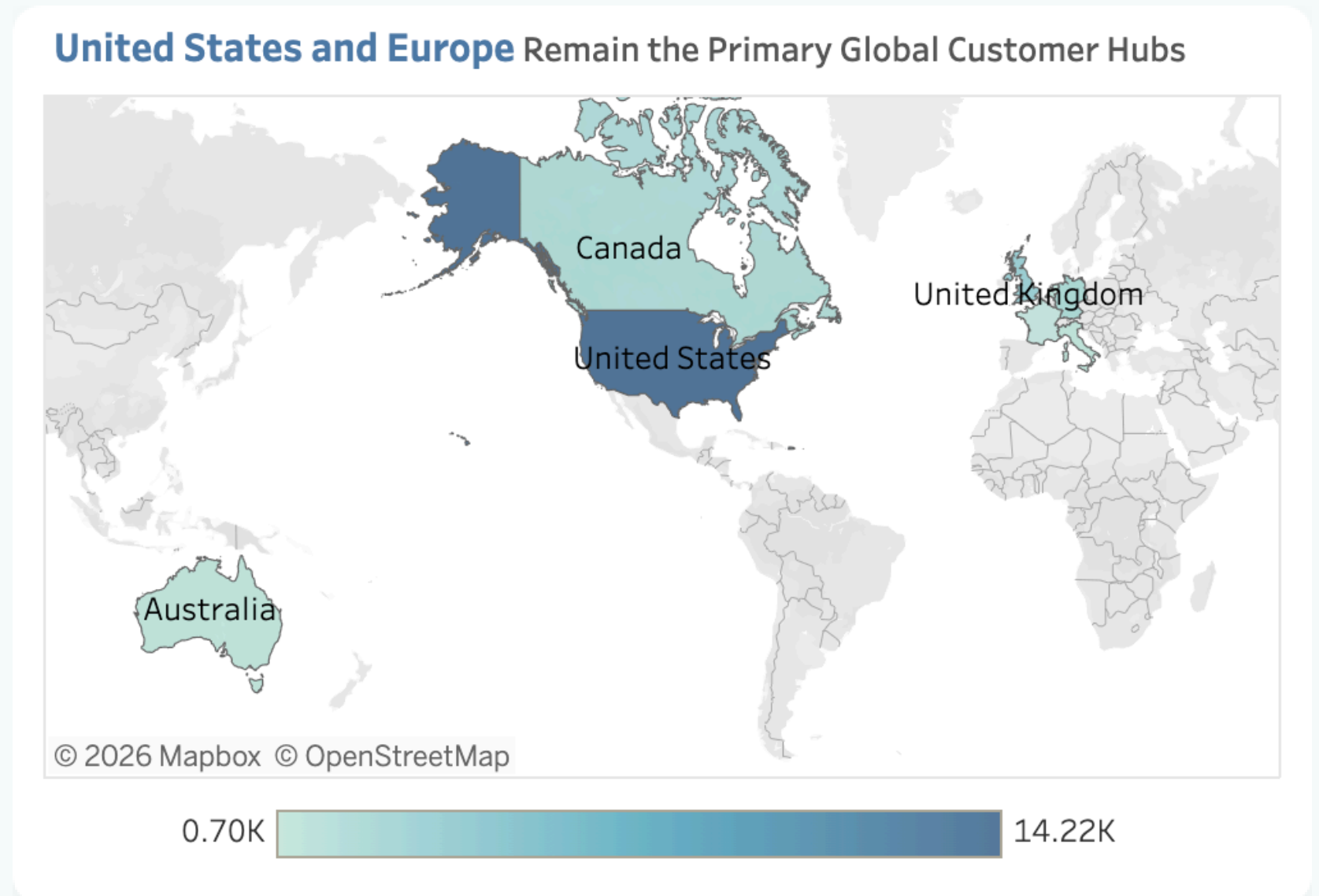
- **Dimensionality Reduction:** Categorized granular age data into distinct macro-demographic buckets (Youth, Adult, Senior).
- **Tech Stack Utilization:** Leveraged Google BigQuery for robust SQL processing, Microsoft Excel for rapid validation, and Tableau for interactive visualizations.

Customer Geography: The Western Anchor

Primary Global Customer Hubs

The geographic distribution of orders reveals a highly concentrated consumer base, essential for targeted marketing and logistics scaling.

- **Market Dominance:** The United States acts as the absolute anchor for the business, generating the highest volume of traffic (reaching up to 14.22K total orders).
- **Secondary Hubs:** Europe, specifically the United Kingdom, acts as the strongest secondary regional hub.
- **Untapped Potential:** Emerging markets like Australia show a growing footprint but currently lack deep market penetration compared to Western hubs.



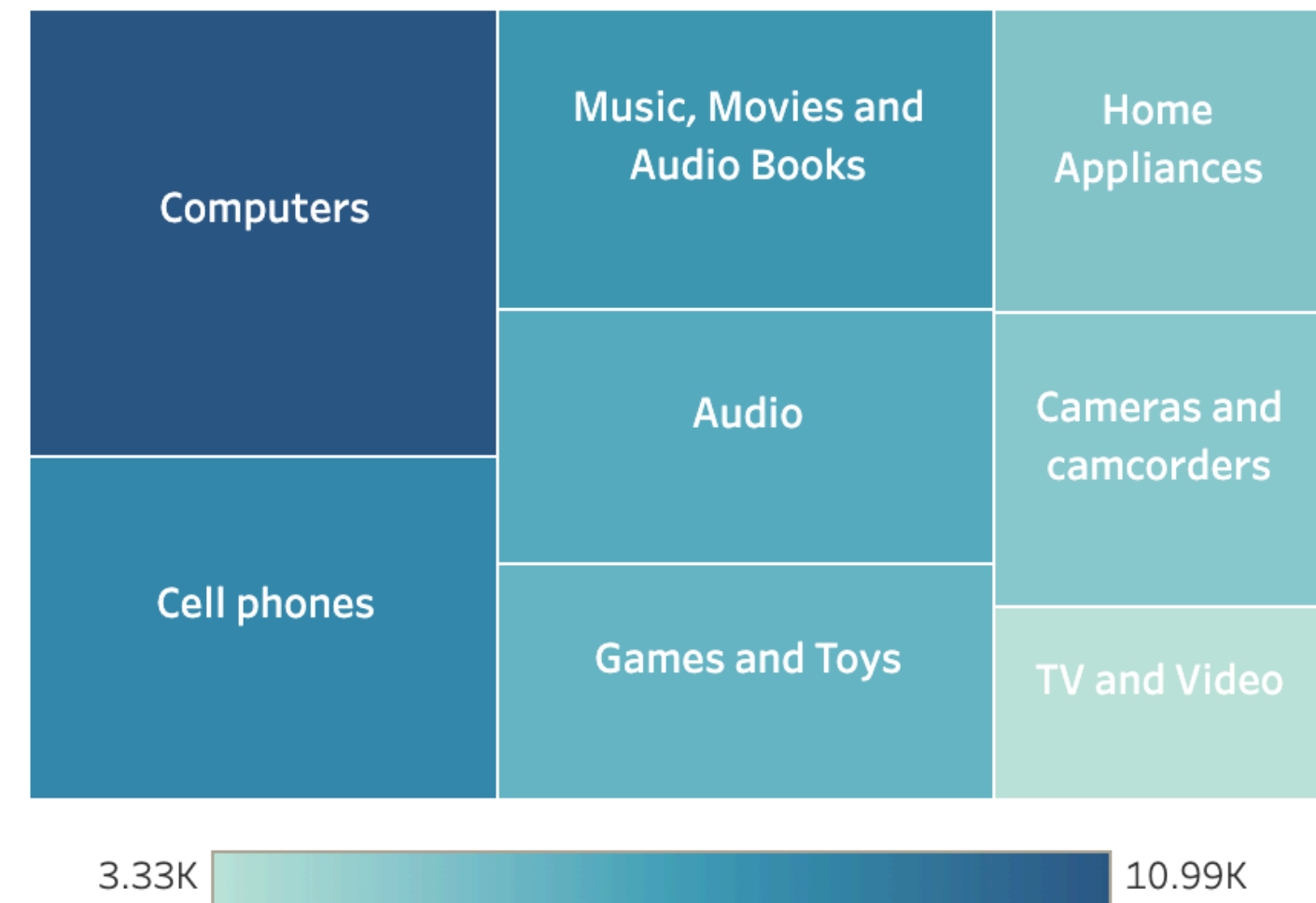
Product Portfolio Dominance

Computers Lead Total Orders

Analyzing the product mix is crucial to understanding where revenue dependencies lie and identifying opportunities for diversification.

- **Category Leaders:** The Computers category heavily dominates the product mix, maxing out the volume tier at nearly 11K orders.
- **Supporting Categories:** It is followed by mid-tier segments such as Cell Phones and Music, Movies & Audio Books.
- **Commercial Takeaway:** The company's current financial stability relies heavily on hardware sales. Diversification is required to mitigate risk.

Computers Lead Product Portfolio by Total Orders



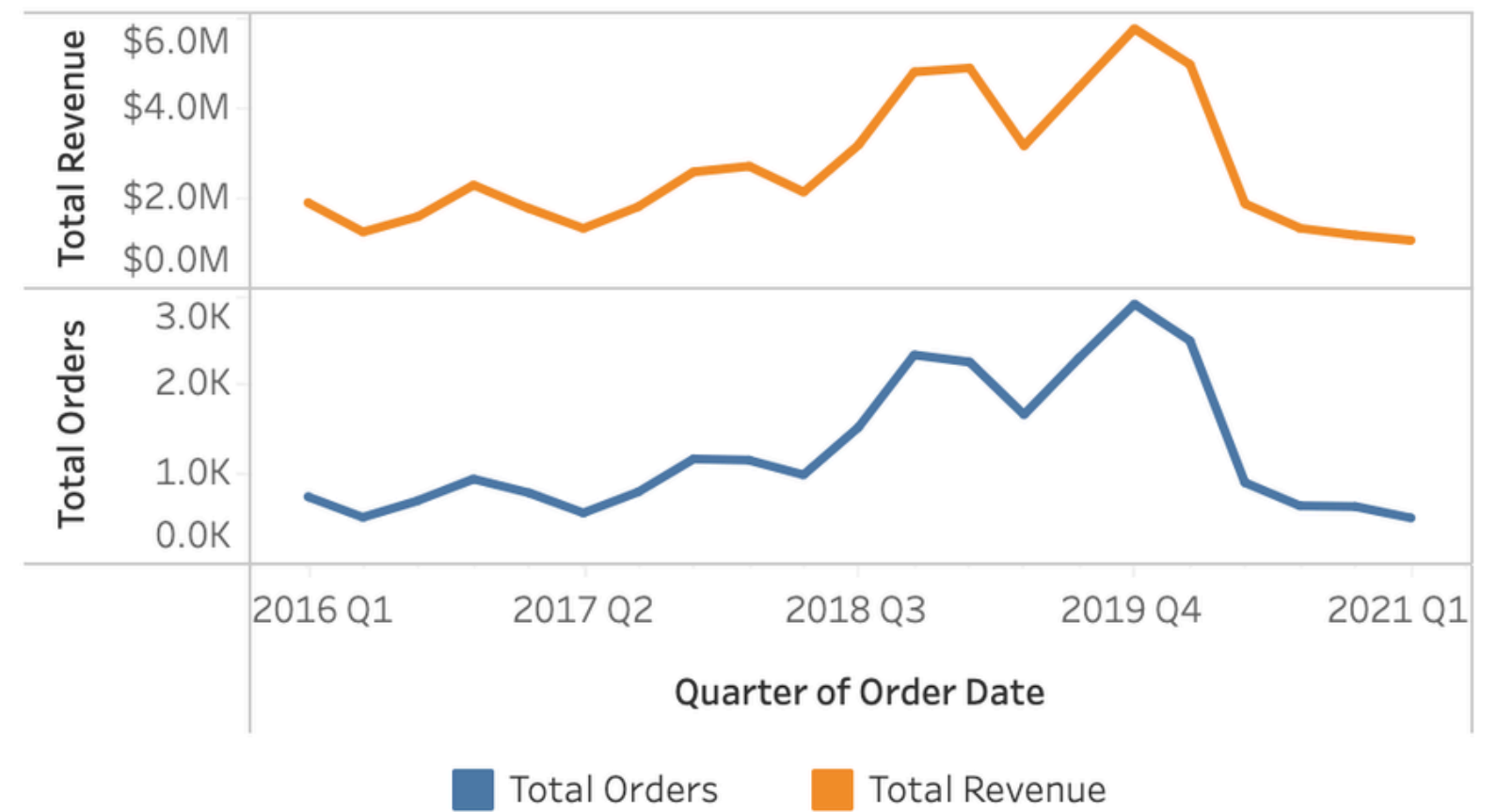
Revenue & Seasonality Trends

The Q4 Holiday Dependency

Historical trend analysis highlights a critical dependency on end-of-year sales, exposing mid-year cash flow vulnerabilities.

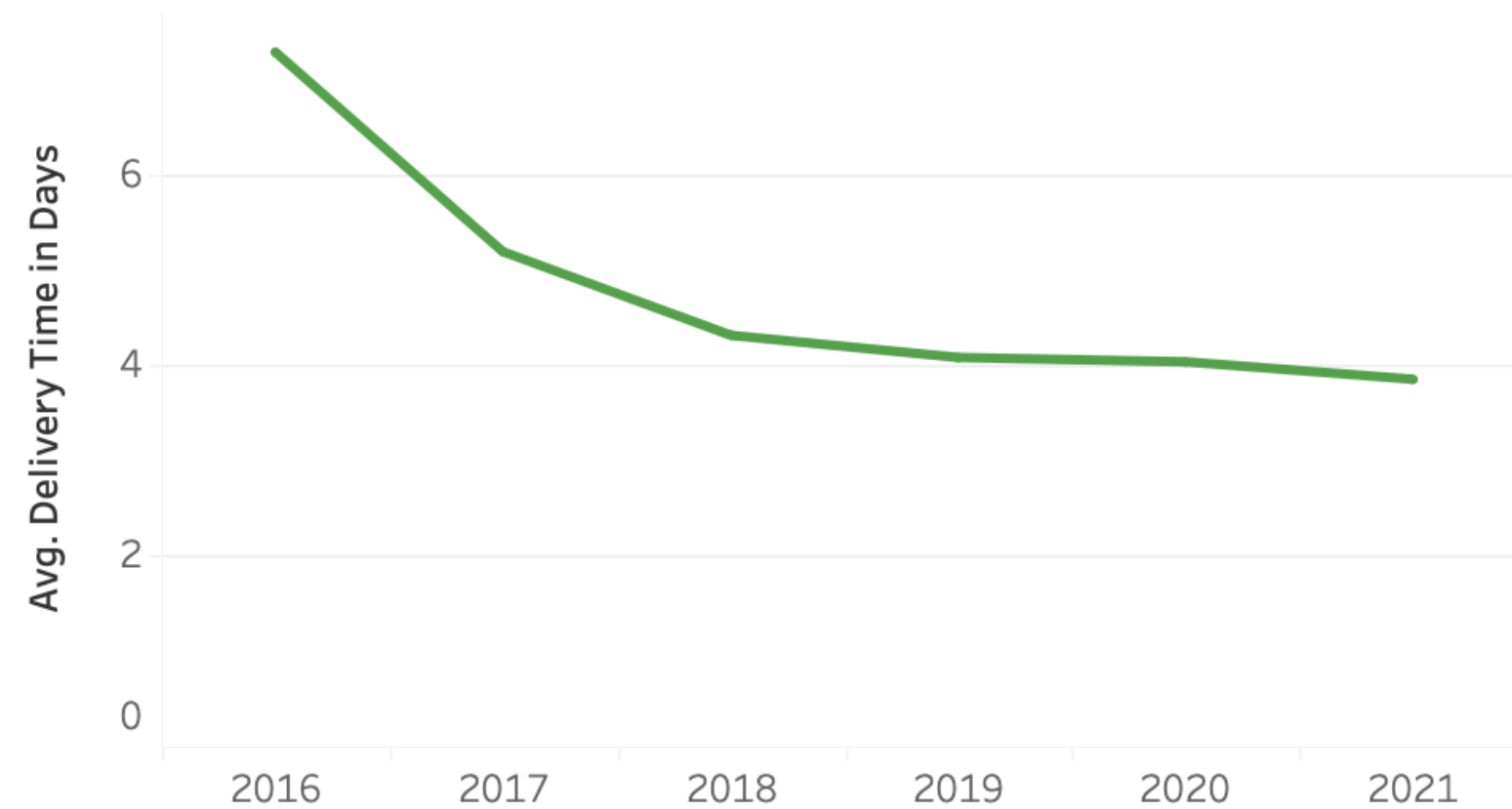
- **Predictable Surges:** There is an aggressive spike in order volume (>3.0K) and total revenue (~\$6.0M) during the fourth quarter of every recorded year.
- **Post-Holiday Drop-off:** The momentum sharply declines moving into Q1 and Q2, reflecting a reliance on the Holiday Season.
- **Commercial Takeaway:** Marketing and supply chain resources are overly stressed during a narrow window, leaving the rest of the year under-optimized.

Q4 Holiday Season Consistently Drives **Peak** Revenue and Order Volume



Operational Efficiency & Channel Performance

Average Delivery Time Remains Stable at **4 Days** Despite Growing Operations



Logistics Optimization

Delivery efficiency has successfully stabilized at a precise 4-day average. The supply chain has proven highly resilient despite growing global volume.

In-Store Sales Generate a Slightly **Higher** Average Order Value Than Online Channels



Channel Value Generation

In-Store transactions consistently yield a higher Average Order Value (\$2.14K) compared to Online (\$2.04K), driven by physical cross-selling advantages.

Strategic Recommendations



Q4 Supply Chain Fortification

Given the reliance on Q4, logistics teams must secure maximum inventory allocation for *Computers* in US and EU distribution centers by late Q3 to mitigate holiday stock-outs.



Closing the AOV Gap

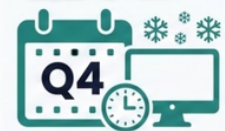
Bridge the deficit between In-Store and Online channels by implementing dynamic bundle pricing algorithms at checkout, replicating the consultative in-store experience.



Off-Peak Market Penetration

Flatten the sharp Q1/Q2 revenue drop-offs by initiating targeted promotional campaigns for mid-tier products during off-peak months, aimed at emerging segments.

Conclusions



This commercial audit confirms that the company's financial backbone is heavily concentrated in Computer sales within Western markets, driven almost exclusively by the Q4 holiday season.



Operationally, the logistics network is highly robust, successfully sustaining a 4-day delivery benchmark across the globe.



By leveraging this stable fulfillment infrastructure to push digital cross-selling and off-peak promotions, the business is well-positioned to evolve from seasonal spikes to sustained, year-round profitability.

Appendix & Resources

DATASET

https://docs.google.com/spreadsheets/d/1aDs0jMjC9sNaI0JgzZlsJsPGH_xBfXezU3XYWsV78Bc/edit?usp=sharing

TABLEAU PUBLIC

https://public.tableau.com/views/GlobalElectronicsRetailerExecutiveCommercialDashboard/ExecutiveDashboard?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

GITHUB

<https://github.com/davidsebastianart/global-electronics-retailer-audit>

Thank You